

CULTURAL FUNDING, ACCESS, AND RESILIENCE

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01/ATTEMPT

I want to use records of NYC’s cultural infrastructure, operating data, demograhpic conditions to explor how different funding decisions could have sustained public opportunities from 2020 to 2025.

Using 2020–2025 keeps the analysis close to documented conditions and allows the simulation to be compared with observed events, budgets, disruptions, facility closures, and program records. A later period such as 2027–2031 could be added as a separate prospective scenario.

02/DATASET

The project requires two linked datasets:

Datasets already available or partially assembled; including facilities, tract boundaries, and selected project-budget information.

Datasets that must still be acquired or constructed; especially program activity, attendance or participation, funding records, travel times, facility operating status, and neighborhood-level demographic and socioeconomic indicators.

02-B/DATASET PRIORITIES

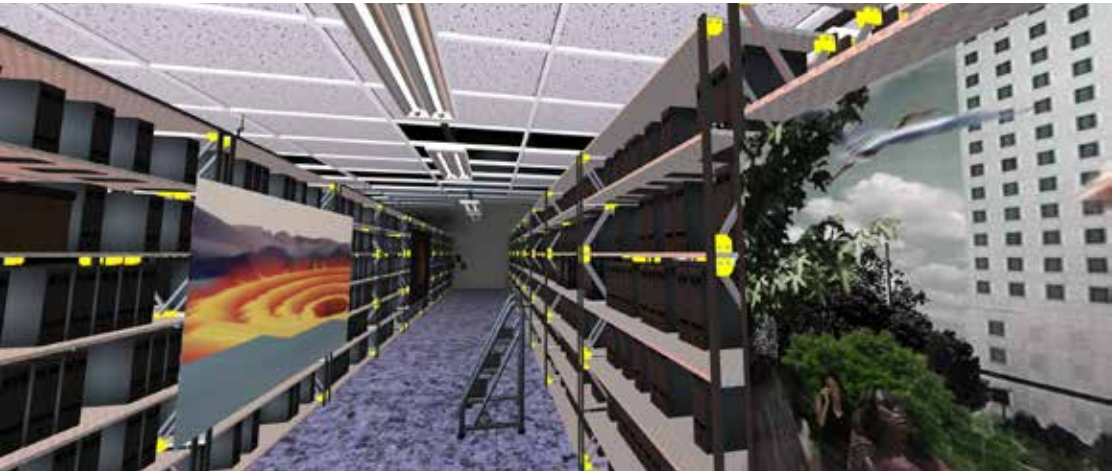
The listed data are organized in their level of order to pursue the project. There are other actors that are in further consideration, such as community & resident inputs, transportation network (physical accessibility), or operating costs and wage data.

03/RESEARCH

a/ Participatory budgeting, a governance precedent: NYC’s The People’s Money as public programming. <https://rnd.council.nyc.gov/ideamap/map/>

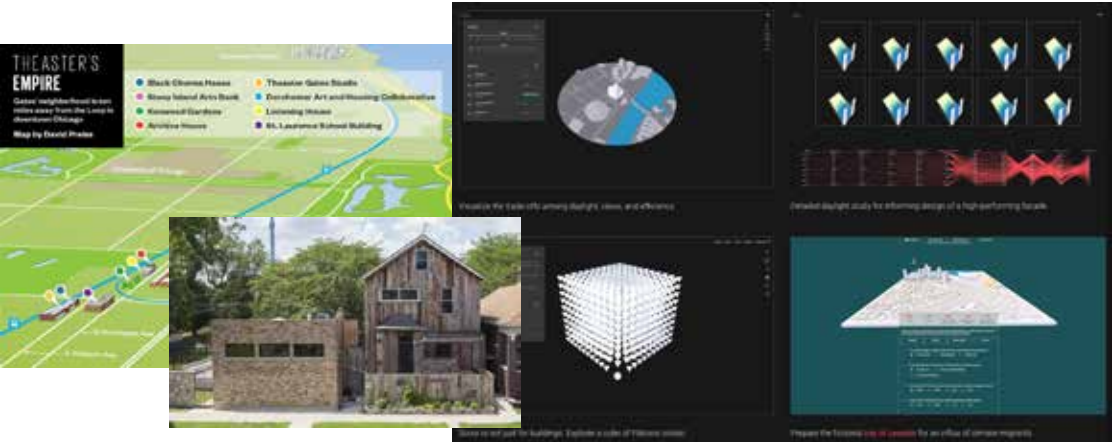
b/ Theaster Gates: precedent for repurposing vacant buildings into public art spaces. <https://www.smithsonianmag.com/innovation/theaster-gates-ingenuity-awards-chicago-180957203/>

c/ KPF’s Scout Build & Leaside City: example of utilizing Pareto Frontier and simulation to create not only critical arguements but also optimizations. <https://scout.build/>



02-A/CORE DATA

dataset	source & purpose	rows & fields needed
Facilities Inventory	NYC Department of City Planning supporting spatial planning	address, lat & long, and categories for 4 cultural types
Census Tracts	US Census Bureau; geographic boundaries for map	GEODID and geometry
Funding & Grant Records	DCLA, city budgets, participatory budgetting	recipients, funder, amount, year, purpose, program
Public Arts Projects	Journals, articles, and published information	facility,annual funding, venue, schedule, project type
Demographic Estimates	US Census Bureau; population and socio-economic conditions	per study area & year, total population, income, access
Digital Infrastructure	comparing and differentiating in times of devastations	platform names, registration, accessibility features

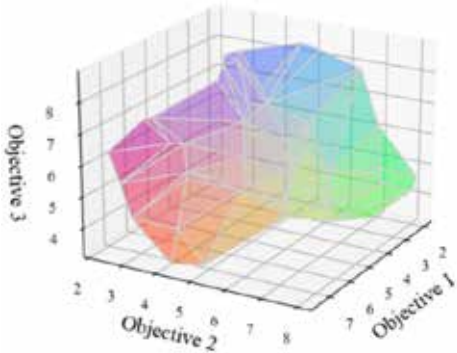




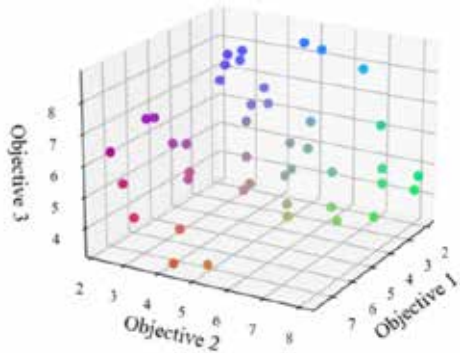
05-A/RULE: Pareto Frontier

The Pareto rule compares complete funding plans and removes plans that another plan clearly outperforms. It does not decide how much equity should be exchanged for more program capacity.

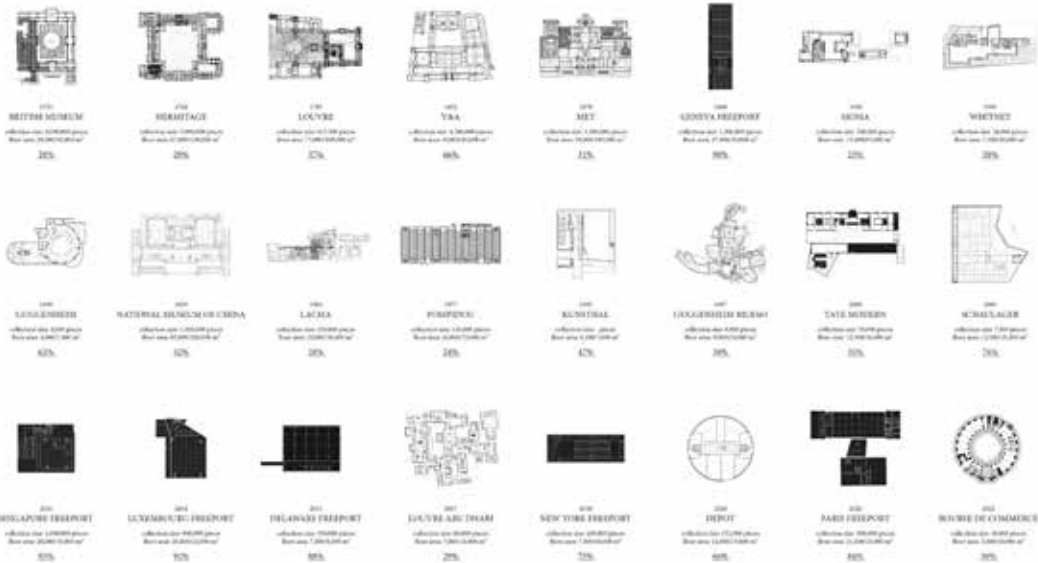
- 1/ Compare it with every other feasible plan.
- 2/ “Does another plan achieve at least as much on both objectives, and more on one?”
- 3/ If yes, remove it from the frontier.
- 4/ If no, keep it as a candidate for discussion.



(a) Pareto front



(b) Vertices on Pareto front



04/RECORDS

project	baseYr	funding	fixCost	costSession	places	maxYr
P01	2020	\$25,000	\$5,000	\$1,000	20	30

05/RULE

At each annual step from 2020 through 2025, update operating costs, determine facility and program availability, calculate how many complete sessions the grant can support, apply observed operating conditions or the disruption scenario where relevant, and redistribute the resulting capacity across the project’s eligible catchment population.



06/MY RUN

Population, locations, and eligibility remain fixed initially unless the data provide reliable annual changes. This isolates funding and operating conditions rather than attempting to forecast the entire city. A later version can compare the historical reconstruction with a prospective 2027–2031 scenario.

07/DATA IT WOULD GENERATE

Tract Shading = Degree of Access or Density
Points = Funded Sites
Point Area = Annual Capacity

Timeline Below:
2020 - 2021 - 2022 - 2023 - 2024 - 2025 → 2027 ~ 2031 (speculate)

Historical comparison: observed program activity, reconstructed capacity, modeled capacity, attendance or registration, and differences by project and year.

Plan summary: cost, six-year capacity, underserved-area access gain, distributional statistics, disruption losses, uncertainty range, validation error, and Pareto status.

Data provenance table: dataset, variable, source, acquisition date, transformation, missingness, and confidence rating.

08/WHAT IT CAN'T SEE

Trust, belonging, discrimination, informal cultural activity / Actual attendance, unique participants, artistic quality / administrative burden, application requirements, and relationships